

Airport Runway Management Simulator Software Development Schedule (10 Weeks)

Project Information

Project Name: Airport Runway Management Simulator

Programming Language: Java

Course: Software Development 1

Week 1: Project Planning and Requirement Analysis

What I Will Do

- **Analyze airport runway management requirements.**
- **Identify system features and modules.**
- **Define project scope and objectives.**
- **Prepare development roadmap.**
- **Research airport traffic management systems.**

What I Need to Learn

- **SDLC**
- **Requirement Analysis**
- **Git & GitHub Basics**
- **Java Project Structure**

GitHub Upload

- **Project Proposal**
- **Requirement Analysis Document**
- **README File**

What I Will Demonstrate

- **Project Idea**
 - **System Features**
 - **Development Plan**
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Week 2: Software Design

What I Will Do

- **Design software architecture.**
- **Create UML Class Diagram.**
- **Design runway workflow.**
- **Plan user interface.**

What I Need to Learn

- **OOP**
- **UML**
- **Java Classes & Objects**

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- **System Design Document**
- **Class Diagram**
- **Project Structure**

What I Will Demonstrate

- **Software Architecture**
- **Workflow Design**

Week 3: User Interface Development

What I Will Do

- **Create main application window.**
- **Design Main Menu.**
- **Add Runway Management options.**
- **Create navigation system.**

What I Need to Learn

- **Java Swing**
- **JFrame**
- **JButton**
- **Event Handling**

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- **Main Menu Source Code**
- **GUI Design Files**

What I Will Demonstrate

- **Working Main Menu**
- **Navigation System**

Week 4: Aircraft Management System

What I Will Do

- **Create aircraft database.**

- **Store aircraft information.**
- **Register landing and takeoff requests.**
- **Manage aircraft records.**

What I Need to Learn

- **ArrayList**
- **Collections Framework**
- **Data Storage**

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- **Aircraft Module**
- **Aircraft Database**

What I Will Demonstrate

- **Aircraft Registration System**
- **Aircraft Management**

Week 5: Runway Allocation System

What I Will Do

- **Assign runways automatically.**
- **Process landing requests.**
- **Process takeoff requests.**
- **Manage runway occupancy.**

What I Need to Learn

- **Queue Data Structure**

- **Conditional Statements**
 - **Event Driven Programming**
 - **GitHub Upload**
 - **Runway Allocation Module**
 - **Request Processing Module**
 - **What I Will Demonstrate**
 - **Runway Assignment**
 - **Request Processing**
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Week 6: Traffic Management System

What I Will Do

- **Create landing queue.**
- **Create takeoff queue.**
- **Handle runway scheduling.**
- **Prevent runway conflicts.**

What I Need to Learn

- **Queue Management**
- **Scheduling Logic**
- **OOP Design**

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- **Traffic Management Module**
- **Queue Processing Module**

What I Will Demonstrate

- **Queue Management**
 - **Traffic Control**
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Week 7: Emergency Handling System

What I Will Do

- **Add emergency landing feature.**
- **Prioritize emergency aircraft.**
- **Improve traffic management.**

What I Need to Learn

- **Priority Queue**
- **Advanced Java Logic**

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- **Emergency Module**
- **Priority Management System**

What I Will Demonstrate

- **Emergency Handling**
 - **Priority Assignment**
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Week 8: Status Monitoring and Reporting

What I Will Do

- **Display runway status.**

- **Show aircraft information.**
- **Generate operation reports.**
- **Create summary page.**

What I Need to Learn

- **GUI Components**
- **Data Presentation**

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- **Monitoring Module**
- **Reporting Module**

What I Will Demonstrate

- **Runway Status Display**
- **Report Generation**

Week 9: Software Finalization

What I Will Do

- **Improve UI design.**
- **Add About Section.**
- **Optimize performance.**
- **Fix minor issues.**

What I Need to Learn

- **Software Testing**
- **UI Improvement**

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- **Updated UI**
- **About Section**

What I Will Demonstrate

- **Complete Workflow**
 - **Improved Interface**
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Week 10: Testing and Final Release

What I Will Do

- **Perform complete testing.**
- **Fix bugs and errors.**
- **Prepare documentation.**
- **Prepare final presentation.**

What I Need to Learn

- **Debugging**
- **Documentation Writing**
- **Presentation Skills**

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- **Final Source Code**
- **Documentation**
- **User Manual**
- **Screenshots**

- **Version 1.0 Release**

What I Will Demonstrate

- **Complete Airport Runway Management Simulator**
 - **All Features Working**
 - **Final Project Presentation**
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Final Features

- **Aircraft Registration**
- **Landing Request Management**
- **Takeoff Request Management**
- **Runway Assignment**
- **Runway Status Monitoring**
- **Landing Queue**
- **Takeoff Queue**
- **Emergency Handling**
- **Traffic Simulation**
- **About Section**

